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Research paper

Topological design of a new family of legged mobile landers based on Truss-Mechanism Transformation method

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ABSTRACT

To meet the needs of the "Returning" and "Base construction" missions of the extraterrestrial body's exploration task, it is necessary to enable the legged stationary lander (LSL) to walk, i.e., it is important and imperative to design novel legged mobile landers (LMLs). This paper aims to address a novel synthesis method based on Truss-Mechanism Transformation (called as TMT method) for LMLs with the capabilities of trusses and mechanisms during different phases. The overall topological design concept and procedure of TMT method are proposed. Based on the extract rules and intersection operations of the motions, the motion requirements for TMT method are obtained. The number condition, the relationship among motions and the methods for allocating the motions in the truss for TMT method are proposed. Some qualitative evaluations are also proposed. By means of this method, numerous structures of LMLs based on the structure of Chang'e lander are synthesized. Finally, after type evaluation, two typical LMLs' legs with transmission limbs or electronic bolts are taken as examples to analyze the properties and abilities of LMLs during different phases. The TMT method is universal and effective for topological design of robots which can transform between truss and mechanism.

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1. Introduction

The project of landing on and exploring extraterrestrial bodies, such as the Moon, Mars, and other asteroids, is an important strategy for some nations to develop the project of deep space exploration. Deep space exploration mainly includes three missions: "Circling," "Landing," and "Returning", while "Base construction" is a mission for the future. So far, the "Circling" and "Landing" missions of Moon have been completed based on the mode which is the combination of stationary lander and a rover [1]. Since the existing landers with legs fixes at the landing site after landing, we call it as a legged stationary landers (LSLs). Furthermore, in this mode, LSLs have two important tasks. One is to help the rover and other devices to soft land on the extraterrestrial bodies' surface, the other is to perform in site exploration at the landing site. Meanwhile, the rover mainly executes major exploration and investigation around the landing site. This exploration mode is suitable to satisfy numerous requirements of the exploration task in a limited range. The design of landers and rovers have gained considerable attention in many nations. The various lunar landers include Surveyor 1 [2], Luna 16 [3], Apollo [4], EuroMoon

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2000 [5], ELENE-B [6], Altair lunar lander [7], Chang'e-3 [8] and so on. The various Mars landers include Viking 1 [9,10] and Phoenix [11]. For rovers, many wheeled rovers were also invented, such as Sojourner [12], Spirit and Opportunity of the Mars Rover mission [13], Jade Rabbit [14], Shrimp and SOLERO [15,16], ExoMars-E [17], ATHLETE [18,19], and Spider-bot [20].

LSLs have the advantages on landing on the extraterrestrial bodies' stability and payload capacity. However, with the development of the extraterrestrial body's exploration task, especially for the task of base construction, it is necessary to let the LSL walk to widen the exploration area. The existing method with a LSL has some limitations, such as: it limits the exploration range of the rover because it requires energy or other aids from the lander after the extravehicular activity; The body of the lander cannot adjust its pose, which is disadvantageous for providing the adequate conditions for the ascent and base construction in the future. Certainly, the proposal of LMLs can meet the requirement of large area exploration, "Returning" mission, and base construction in the future. Thus, it is critical and significant to study the research on LMLs especially for the novel design of their mechanisms. Some novel exploration modes should be proposed to overcome the previously mentioned limitations.

So far, a few studies have discussed some novel methods or exploration strategies for exploring the extraterrestrial bodies. Zhang et al. [21] proposed the conceptual design of a manned lunar lander with wheel-legged mobile system, which enhances the capability of lunar exploration. Nurahmi et al. [22] introduces a methodology for the type synthesis of two degree-of-freedom hybrid translational manipulators. In our previous work [23], we presented a novel exploration mode for the explorative task. The lander has the multiple abilities of stationary landers and rovers with landing and walking functions. Furthermore, we also designed this type of LMLs with a passive limb [23]. In addition, some special LMLs were obtained with a rhombus joint by the Lie group and differential manifold methods [24]. The main backbone needs to bear most of the impact force during landing; thus, one actuated limb (also called transmission limb) is used to protect the actuated joint against the impact force. Numerous structures of the LMLs were proposed. However, there are numerous tasks that still need to be addressed especially for the topological design of their mechanisms.

Essentially, designing legged landers largely depends on the corresponding topological synthesis method. As for LSLs, there is no reference to report the topological design method of stationary landers, whose main functions include landing and being deployable. Since the function of being deployable needs only one degree of freedom, the structure can be designed by intuition directly. However, as for LMLs, its functions are multiple and complex, which include being deployable, landing buffer, walking, orientation adjustment, and terrain adaptability functions. The corresponding motions of LMLs are complicated, and one of the biggest issues is the degree of freedom (DoF) contradiction: the configuration of LMLs during landing phase needs to be stationary without any motion or DoF, while the configuration during walking phase needs multiple motions or DoFs. In our previous work [23,24], some novel structures of LMLs were proposed based on the type synthesis method [25–30] without considering the existing structures of stationary landers. To satisfy the requirements of the exploration mission for the aerospace department, it is meaningful and practical to design structures of LMLs based on the structure of the existing stationary lander. Furthermore, in order to overcome the DoF contradiction of LMLs during different phases, it is significant to study a methodology to design novel kinematic architectures of LMLs based on Truss-Mechanism Transformation.

However, to the best of our knowledge, there is no literature that discusses the TMT method for designing LMLs systematically. The method for designing a robot with truss and mechanism functions (which has landing and walking functions) still need to be further studied. In this study, we design innovative LMLs with multiple functions based on the TMT method. Some of its advantages are as follows. The structures of landers enhance the exploration capability as compared to the combination model of stationary lander and rover with lower weight and cost. In terms of design approach or concept, we proposed a concept for designing a novel mechanism based on the TMT method and Chang'e structure, which is an effective method to design a robot with the capability of a truss and a mechanism during different phases.

The main objective of this study is to investigate the topological synthesis method (TMT method) of a special family of LMLs based on Truss-Mechanism Transformation. The remainder of this study is organized as follows. In Section 2, we outline the overall concept and procedure of TMT method. We present the TMT method in detail in Section 3. In Section 4, we design the structures of legs based on TMT method and Chang'e structure. In Section 5, two typical LMLs' legs with transmission limbs or electronic bolts are taken as examples to analyze the properties and abilities of LMLs during different phases. Finally, we conclude this study in Section 6.

2. Overall topological design concept and procedure of TMT method for LMLs

2.1. Task requirements of LMLs

As mentioned in the published articles [23,24], a novel LML should accomplish five functions: being deployable, landing buffer, walking, orientation adjustment, and terrain adaptability functions. It is noted that DoF of the mechanism during landing and walking phases is one of the biggest challenges: during the landing phase, the LML can be considered as a truss or structure (without any motion); during the walking phase, it can be considered as a multi-DoF robot (with some motions). Therefore, the transformation of LMLs from the landing position to the walking position can be seen as that from a truss to a mechanism.

Therefore, this problem translates to other problems, such as how to design a mechanism based on a truss, how to allocate the motions into the given truss, and how to transform the positions of a mechanism from truss to mechanism.

2.2. Novel design concept of TMT method

As for this type of special robot with the properties of truss and mechanism, the common type synthesis approaches cannot be applied directly. Considering that a truss can bear heavy load or large impact, the structure of an LML in the landing phase can be seen as a truss. Meanwhile, during the walking phase, the structure of an LML is a mechanism that has the flexible ability for walking. To some extent, some phenomena are concluded: (1) the topological arrangements or graphs of a truss and the corresponding mechanism located at its particular position without motion can be the same. (2) The positions or configurations of a mechanism and its corresponding truss can be transferred. (3) The locations of the nodes and the links of the truss determine the candidate positions of the joints, which will be added into the truss. Based on these phenomena, a truss is considered as a mechanism which is in its particular position without motions, while a mechanism is seen as a truss by adding some motions in its nodes and links. There are some methods for mechanisms to maintain the particular positions without motion of truss (this position is also called as Truss Position of Mechanism, TPM), which will be illustrated in Section 3.3. Therefore, one novel design concept is proposed to be an effective way to design LMLs with the multiple functions of truss and mechanism based on TMT.

2.3. Synthesis procedure of LMLs based on TMT method

According to the above concept, this paper will develop the TMT method to obtain new structures of LMLs. The general procedure of the TMT method for LMLs is organized as shown in Fig. 1. It contains four stages with several steps. Stage 1: get the motion requirements by extracting the motions from the functions or tasks. Stage 2: design trusses for the task. Stage 3: design the mechanism based on the achieved truss and motion requirements. Stage 4: perform type evaluation of the corresponding robots according to the indexes. Finally, through this procedure of TMT method, the desired robot can be obtained. The design process of TMT method will be developed in the subsequent sections.

3. TMT method for topological design of LMLs

3.1. Motion requirements for TMT method

In topologic synthesis of the mechanism, the first important issue is to define the task and extract its mapping motions. To facilitate the extraction and presentation of the motions, some rules, geometric and symbolic expressions are provided. As mentioned in Stage 1, the extraction rules are as follows:

Rule 1 (for rotation motion): if the end-effector can rotate around a certain axis $\boldsymbol{u}$, a rotation motion (R-motion) will be required whose axis coincides with the aforementioned axis. Based on Herve's work [27], the corresponding displacement subgroup is $\{R(N, u)\}$.

Rule 2 (for translation motion): if the end-effector can translate in a certain direction $\boldsymbol{u}$, a translation motion (T-motion) will be required with a direction aligned in the same direction, whose displacement subgroup is $\{T(u)\}$.

Note that more motions are achieved by the method formed from the above two rules. Furthermore, the mapping motions can be represented as a subgroup or submanifold, which is a composition obtained by implementing the product of all displacement subgroups. Furthermore, some geometrical symbols are listed in Table 1 to represent the motions intuitively.

After obtaining the motions with respective to each function, the integration of the motions need to be carried out. Thus, it is necessary to put forward the intersection rules or operations of motions. In the topologic synthesis of parallel manipulators, the intersection operations of the Lie subgroup or submanifolds is an important criterion. Here, we list the basic rules of the intersection operations (which is not null) among one-dimensional (1-D) motions.

(a) If axes of two rotational motion could coincide with each other, the intersection of them is 1-D rotational motion whose axis coincide with the given axis. Furthermore, there are three cases which are appropriate for this situation:

$$\begin{cases} \{R(A,\boldsymbol{u})\} \cap \{R(B,\boldsymbol{u})\} = \{R(N,\boldsymbol{u})\} N \in l_{AB}, \boldsymbol{u}=\vec{AB} \\ \{R(A,\boldsymbol{u})\} \cap \{R(\boldsymbol{u})\} = \{R(A,\boldsymbol{u})\} \\ \{R(\boldsymbol{u})\} \cap \{R(\boldsymbol{u})\} = \{R(\boldsymbol{u})\} \end{cases} \tag{1}$$

Where $\{R(A, \boldsymbol{u})\}$ presents one dimension (1-D) rotation about axis parallel to unit vector $\boldsymbol{u}$ and passing through point A. $\{R(\boldsymbol{u})\}$ means 1-D rotation about axis parallel to unit vector $\boldsymbol{u}$ with infinite axis. l_{AB} means the line passing through points A and B. $N \in l_{AB}$ means the point N is located in line AB.

(b) For translational motions: if there exists a T-motion along with the two directions of two T-motions, whose direction is parallel to their directions, it can be expressed as follows:

$$\{T(\boldsymbol{u})\} \cap \{T(\boldsymbol{u})\}=\{T(\boldsymbol{u})\} \tag{2}$$

Where $\{T(\boldsymbol{u})\}$ donates 1-D translational motion whose direction parallel to the unit vector $\boldsymbol{u}$.

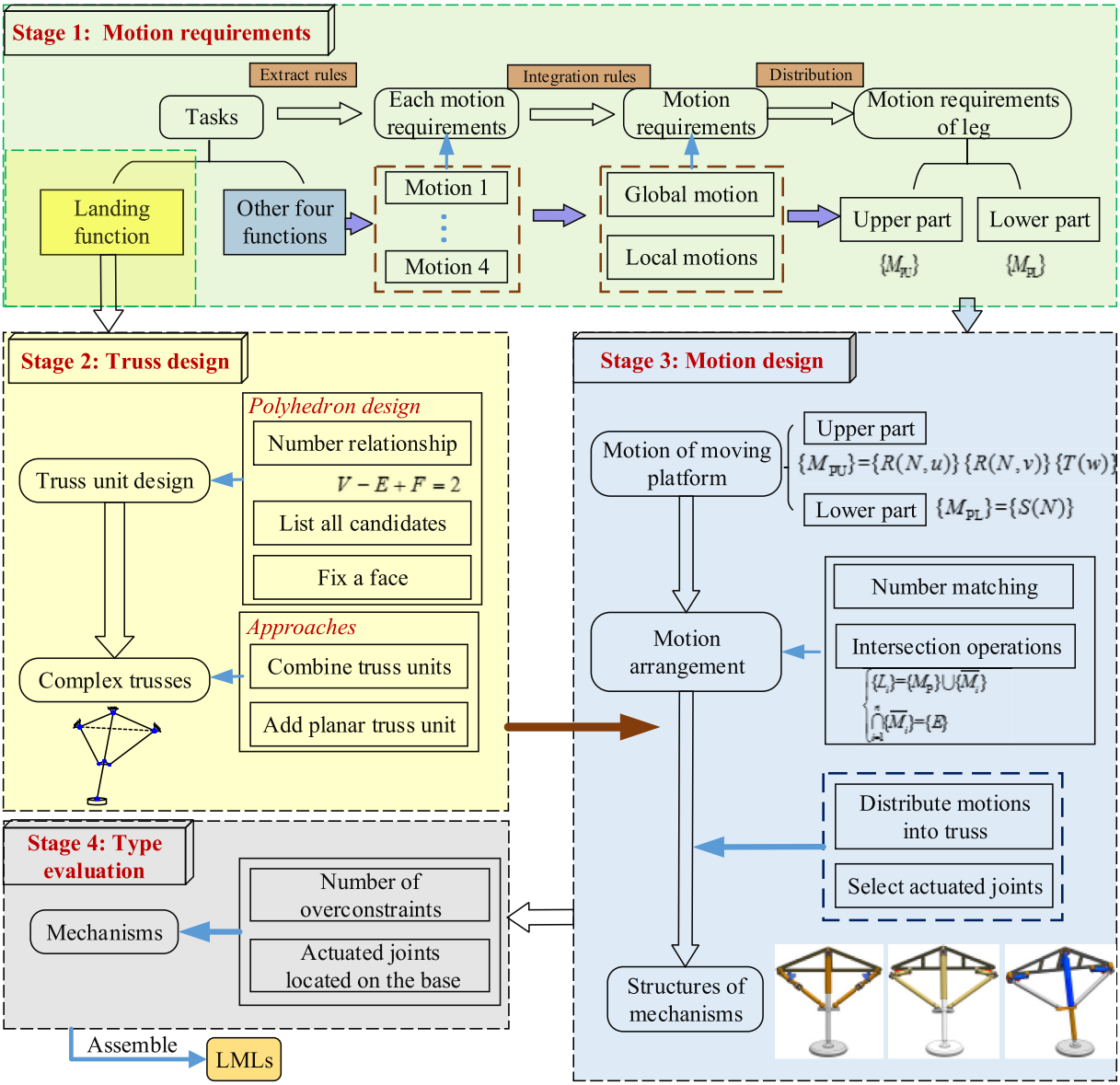


Fig. 1. Synthesis procedure of TMT method.

Table 1
Symbols definition and explanation [27].

Symbol	Subgroup	Motion
–	$\{E\}$	Rigid connection without relative motion
	$\{R(N, \boldsymbol{u})\}$	1-D rotation about axis parallel to unit vector $\boldsymbol{u}$ and passing through point N
	$\{R(\boldsymbol{u})\}$	1-D rotation around any axis which is parallel to unit vector $\boldsymbol{u}$, called rotation with infinite axis
	$\{S(N)\}$	3-D rotation about point N
	$\{T(\boldsymbol{u})\}$	1-D translation parallel to the unit vector $\boldsymbol{u}$
	$\{T(P_{vw})\}$	2-D translations within plane P_{vw} formed by two linearly independent unit vectors $\boldsymbol{v}$ and $\boldsymbol{w}$
	3T	3-D translations in space
	$\{C(N, u)\}$	Cylindrical motion about the axis $(N, \boldsymbol{u})$
	$\{D\}$	6 rigid body motion

Furthermore, it is important to note that the higher dimensional motions can be spanned by the 1D motions, such as 3D spherical motion can be spanned by three R-motions whose directions intersect at a point. Thus, more principles for obtaining the intersection of any two complex motions can be obtained and represented based on the basis rules.

3.2. Truss design for TMT method

Topologic synthesis of mechanisms based on TMT method needs to achieve a truss before considering the motion design. Followed by stage 2, a truss is formed by straight links connected by nodes. A space frame truss is a 3-D framework of members that are pinned at their ends. A space truss unit is polyhedral in shape. From this perspective, we can design a space truss from a polyhedron under some conditions detailed as follows:

Step 1: Determine the numbers of vertices (V), edges (E), and faces (F) of a polyhedron according the relationship among them based on the Euler's polyhedron formula, which is expressed as follows:

$$V - E + F = 2 \tag{3}$$

Step 2: List all the possible candidate polyhedrons based on the numbers of V, E, and F.

Step 3: Select a face of the polyhedron, and fix it to be a base platform of a mechanism. Accordingly, the edges are seen as limbs of mechanism, and the numbers of edges (EL) and limbs are equal.

Step 4: Complex trusses are obtained following the two methods detailed as below. (1) Combine the designed truss unit with a planar truss unit, such as adding two links connected by a hinge joint to a rigid of the original truss with hinge joints on the both sides, which could not change the force condition. (2) Assemble two or more designed truss units reasonably. After getting the complex truss, some more nodes should be fixed if necessary.

3.3. Motion design for TMT method

The motion design of mechanisms (which is the third stage) needs to satisfy two main conditions: one is the number condition including the relationship among DoF, number of active joints, passive joints, etc.; the other is relationship among detailed properties of joints. As for the TMT method, it is necessary to carry out the approaches to allocate motions or replace the motions into the nodes and links of truss. Finally, the actuated joints need to be selected.

3.3.1. With respect to the number condition

For TMT method, it needs to consider not only the conventional numbers relationship, but also the number of edges of truss, which is described as below.

The DoF of a mechanism can be established by the modified G-K criterion [30], which is as follows:

$$\begin{cases} M_{DoF} = d(n-g-1) + \sum_{i=1}^{g} f_i + \nu - \xi \\ d = 6 - \lambda \end{cases} \tag{4}$$

where M_{DoF} is the degree of freedom of mechanism, d denotes the order of the mechanism, λ is the number of the common constraints of the mechanism, n is the number of links, g is the number joints or pairs, f_i is the number of DoF of the ith joint, v is the number of redundant constraint, and ξ is the passive DoF.

A parallel mechanism is a multi-loop mechanism, wherein the number of the loop can be obtained as follows:

$$l = g - n + 1 \tag{5}$$

where l denotes the number of the loop.

Substituting Eq. (5) into Eq. (4),

$$\sum_{i=1}^{g} f_i = M + dl - \nu + \xi \tag{6}$$

Where M is the sum of dimension of the independent moving platform's motion.

Further, it can be obtained as follows:

$$\sum_{j=1}^{n_{TL}} C_j = \sum_{i=1}^{g} f_i \tag{7}$$

where C_j is the DoF in the jth limb.

According to Eqs. (6) and (7), it can be obtained as follows:

$$\sum_{j=1}^{n_{TL}} C_j = M + dl - \nu + \xi \tag{8}$$

Based on the above equations, C_j can be determined after identifying M, d, l, v, n_{TL}, and ξ.

Furthermore, once the dimension number of the end-effector's motion is identified, the relationships between the number of limbs, number of actuated limbs, number of actuators of limbs, and number of passive limbs are as follows:

$$\begin{cases} n_{\mathrm{TL}} = n_{\mathrm{PL}} + n_{\mathrm{AL}} \\ n_{\mathrm{AL}} = A_C - \sum_{i=1}^{n} (A_{Ci} - 1) \\ A_C = R_A + M \\ n_{\mathrm{TL}} = E_L \end{cases} \tag{9}$$

where n_{TL} is the number of total limbs; n_{AL} is the number of limbs with input actuators; n_{PL} is the number of passive limbs (limbs without input actuators); A_C is the total number of active joints of the mechanism; A_{Ci} is the input number of limb i; R_A is the number of the redundant actuator for actuator redundancy; E_{L} is the number of edges of truss, which is used as limbs for mechanisms.

3.3.2. With respect to the relationship among motions or joints

The existing type synthesis methods mainly focus on the degree of freedom and directions of motions without considering their detailed positions. However, as for TMT method, in order to accomplish the distribution of the motion in the nodes and links of a truss, it is necessary to take the detailed positions into account. Thus, this paper proposes a method to design the motion of each limb considering the positions of motions according to the motion of the moving platform and DoF.

Firstly, the relationship of motions between the moving platform and the limbs, or the composition principle of mechanisms, needs to be described. Based on the Lie group theory [27,28], the displacement subgroup of the moving platform for parallel mechanism is the intersection of the subgroup or submanifolds produced by all limbs, which can be expressed as follows:

$$\{M_{\mathrm{P}}\} = \bigcap_{i=1}^{n} \{L_i\} \tag{10}$$

where $\{L_i\}$ is the displacement subgroup or submanifold of the ith limb where $i = 1, 2, 3 \ldots$, $\{M_{\mathrm{P}}\}$ is the displacement submanifold of the moving platform. $\cap$ is a set intersection operator. For example, $A \cap B$ represents the mechanisms A and B connected with each other in a parallel form.

The displacement or submanifolds of each limb must satisfy the following equations:

$$\begin{cases} \{L_i\} = \{M_{\mathrm{P}}\} \cup \{\overline{M_i}\}, i = 1, 2, 3 \ldots \\ \cap_{i=1}^{n} \{\overline{L_i}\} = \{E\}, i = 1, 2, 3 \ldots \end{cases} \tag{11}$$

where $\{\overline{L_i}\}$ is the additional displacement subgroup or submanifold of ith limb except the target submanifold $\{M_{\mathrm{P}}\}$. $\{E\}$ is the rigid connection without relative motions.

Secondly, the detailed procedure to determine the motions is explained in Fig. 2, described as following:

(1) Number synthesis. The relationships among fundamental topology elements of mechanisms can be achieved according to Eqs. (8) and (9), including the number of limbs, number of actuated limbs, number of actuators in a limb, and DoF of each limb.
(2) Add the motion of the moving platform to each limb directly. These motions are necessary because $\{M_{\mathrm{P}}\} \in \{L_i\}$ According to Eq. (10). As for R-motion, it has two approaches: one is to add the R-motion whose axis is coincide the desired R-motion; the other one is to add a R-motion which has the movability [23,27–29] and its axis is parallel to that of the desired R-motion. In the second approach, it introduces 2-D T-motion. Thus, it also needs to satisfy the condition$\cap_{i=1}^{n} \{\overline{L_i T}\} = \{E\}$, where $\{\overline{L_i T}\}$ means the additional translational displacement subgroup or submanifold of ith limb except the target translational submanifold. As for T-motion, the direction of the added T-motion must parallel to that of desired T-motion.
(3) Add more motions except the motion of moving platform, by checking the condition of Eq. (11) by applying the intersection rule of Eqs. (2) and (3). As for R-motion, it needs to satisfy the condition $\cap_{i=1}^{n} \{\overline{L_i R}\} = \{E\}$, where $\{\overline{L_i R}\}$ means the additional rotational displacement subgroup or submanifold of ith limb except the target rotational submanifold. As for T-motion, it needs to satisfy the condition $\cap_{i=1}^{n} \{\overline{L_i T}\} = \{E\}$.

3.3.3. With respect to the approaches to allocate the motions in the truss for TMT method

To design a mechanism based on the truss, it is necessary to add some joints into the node or the links and provide the methods for mechanisms to maintain TPM. Two methods to maintain TPM of a mechanism is explained as follows:

Dead Point [32] Based TPM (**TPM-DPB**): One type of TPM is obtained when the axes of some links are coincident to reach its dead point which a kind of singularity position. Considering the planar four-link mechanism as an example, when the axes of the driving link rocker and the passive link crank are coincident, the crank cannot move no matter how large the moment attached on the driving link is on this stationary position. This singularity position is also called a dead point of the mechanism.

Fixed Joint Based TPM (**TPM-FJB**): One type of TPM can be obtained when some links reach its limiting position, such as: for two links connected by the P joints, the links cannot translate in one direction if reaching the extreme position; for two links connected by the R joint, one link cannot rotate if it is locked by the added locking device.

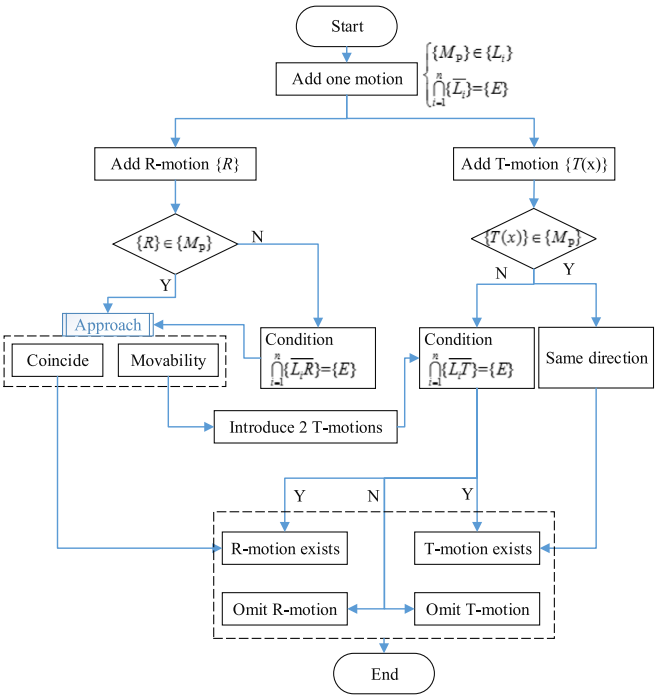


Fig. 2. Procedure of obtaining the motions of limbs.

In the following, we obtain all the possible topological structures of limbs and determine the joint configurations on the truss according to the motion of limbs. There are mainly two approaches: one is to increase the number of joints; the other one is to obtain equivalent joints, which is explained in detail as follows:

(1) For increasing the number of joints: It is easy to implement two approaches. One is to add a 1D or complex rotational joints to the node to obtain the R-motions, the other one is to add a prismatic or rotational joint to a link to gain more T-motions or R-motions according to the two methods (TPM-DPB and TPM-FJB). To maintain the maximum capability of the singularity configuration, it is supposed that only one joint is added in one link of the truss.
(2) For the equivalent joints: Composite joints are employed to replace the combination of simple joints and some other equivalent joints are used to replace the original joint, such as using the parallelogram joint to replace the prismatic joint.

Based on the above methods, the following topological structures of legs are obtained: (1) starting from some simple and institutive kinematic limbs with the required motions of limbs. (2) According to the nodes and links on the edges of truss, employ some complex joints to replace the simple joint if necessary. (3) Based on the aforementioned approaches, distribute the joints in the nodes and links of the truss. (4) Obtain more limbs of mechanisms by using the equivalent joints. (5) Arrange the joints in the truss.

3.3.4. With respect to the selection of actuated joints

The actuated joints could not be selected arbitrarily [31]. The selection of the actuated joints must ensure that, in a general configuration, the DOFs of the mechanism with all the blocked actuated joints is zero.

3.4. Type evaluation indexes

Finally, in the fourth stage evaluation indexes carried out, which contains include qualitative and quantitative evaluation indexes. However, we mainly focus on the novel structure design of LMLs without considering the detailed parameters; thus, it focuses only on the qualitative index. Hereby, two criterions are considered.

Criterion 1: Number of overconstraints;

The mechanisms with overconstraints have some features. First, it may form a mechanism more complex than a mechanism without overconstraints. Furthermore, this type of mechanism requires high manufacturing precision and assembly precision, otherwise some links cannot be installed. Finally, it easily forms the internal force of the mechanism. If the mechanism undergoes a certain deformation after being subjected to heavy load, it may change the geometric constraint of the mechanism, and its mechanism cannot perform the expected motion or even any movement. For the above-mentioned

reasons, it is believed that a smaller number of overconstraints of mechanisms makes the mechanisms simpler, and its size optimization becomes simpler.

The number of overconstraints of the mechanism is obtained as follows:

$$\Delta = \sum_{i=1}^{m} c^i - c \tag{12}$$

where Δ is the number of overconstraints if $\Delta > 0$. c is the order of the wrench system of the parallel mechanism's platform, m is the total number of limbs, and c^i is the order of the wrench system of limb i. In terms of motion, Eq. (12) can be deduced as follows:

$$\Delta = \sum_{i=1}^{m} [6 - D(L_i)] - (6 - D(M_i)) = 6(m-1) - \sum_{i=1}^{m} D(L_i) + D(M_i) \tag{13}$$

where $D(L_i)$ and $D(M_i)$ are the dimension of motions of limbs and the moving platform of the mechanism, respectively.

Criterion 2: Number of actuated joints located on the base;

It is known that if the actuator is located on the base, it will decrease the load and moment of inertia. With a given number of actuated joints, a robot exhibits better kinematic and dynamic performances with more actuated joints located on the base. Therefore, the number of actuated joints is taken as an index, which is expressed as follows:

$$\eta = \frac{\sum_{i=1}^{m} J_A^{Base}}{\sum J_A} \tag{14}$$

where η is the proportion of the number of actuators located on the base $\sum_{i=1}^{m} J_A^{Base}$ to the total number of actuators ΣJ_A.

Based on the above-mentioned criteria, the mechanisms with better performance can be selected.

4. Innovative design of legs for LMLs based on TMT method

4.1. Motion requirements of legs

Based on the displacement subgroup method [25,26] and the rules described in Section 3.1, the motion requirements of the leg is extracted and integrated, which is represented as the subgroup $\{D\}$. $\{D\}$ is the result of the implementation of the composition product of two subgroups, $\{T\}$ and $\{S(M)\}$, i.e., $\{D\}=\{T\}\{S(M)\}$, where M is located on the center of the footpad. According to two previous studies [22,23], the leg of the mobile lander consists of two parts, the upper and lower parts. The subgroup of the lower part is $\{M_{\mathrm{PL}}\}=\{S(M)\}$. The subgroup of the point of the upper part (or the center of the footpad) is $\{M_{\mathrm{PU}}\}=\{T\}$. Furthermore, the upper part is regarded as a parallel mechanism whose point of the moving platform (called as mid-platform) allows 3D translations-$\{T\}$. The motions of the point attached on the footpad can be realized by other motion distributions of mid-platform. To realize the center of the footpad's 3D translations, the mid-platform's motions could be represented as eight submanifolds by means of enumeration methods [24], which could be proven by the subgroup method. For reasons of space and clarity, four motions which are suitable for designing the legs are chosen to be designed, including $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$, $\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\{R(N_2,\boldsymbol{v})\}$, $\{R(N,\boldsymbol{v}_1)\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$ and $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$, as shown in Fig. 3 in this paper. In addition, other cases can be performed by means of the same method.

4.2. Truss design of legs

The structure of Chang'e lander [2] is composed of main body and four legs, and each leg contains the same structures with three backbones: one main backbone and two auxiliary backbones. Three backbones connect with the body by the U joints, while the auxiliary backbones connect with the main backbone by S joints. The structure of each leg consists of two parts: upper part and lower part: The upper part is a parallel structure and the lower part has the terrain adaptability mechanism connected with a foot pad by a passive S joint. According to the structure topology of Chang'e lander and the procedure of the topology design of space truss, we select V=6, E=9, and F=5. Thus, one polyhedron $ABCEFG$ of the upper part of the leg is obtained. Then the link AE is extended up to the center M of the foot pad, one truss topology of the legs for LMLs is obtained, as shown in Fig. 4. The face ABC is chosen to be fixed to be the base platform and the face EFG is selected as the moving platform (here also called as Mid-platform). Then E_{L} equals 3.

4.3. Motion design of legs

Since the leg's structure consists of upper part and lower part, each of them is designed as following.

As for the lower part: The motion of the lower part of the leg is represented as subgroup $\{S(M)\}$. According to the Lie group theory, many different structures for this displacement subgroup $\{S(M)\}$ could be achieved, such as: $({}^{u}R^{v}R^{w}R)_M$, ${}^{uv}U_M{}^{w}R_M$, ${}^{u}R_M{}^{vw}U_M$, and S_M. Here, ${}^{uv}U$ is a universal joint, which rotates around the axes $\boldsymbol{u}$ and $\boldsymbol{v}$. S_M is the spherical joint, which rotates around the point M. For simplicity, only S_M joint is chosen to determine its subgroup, which locates on the center of the footpad in this study.

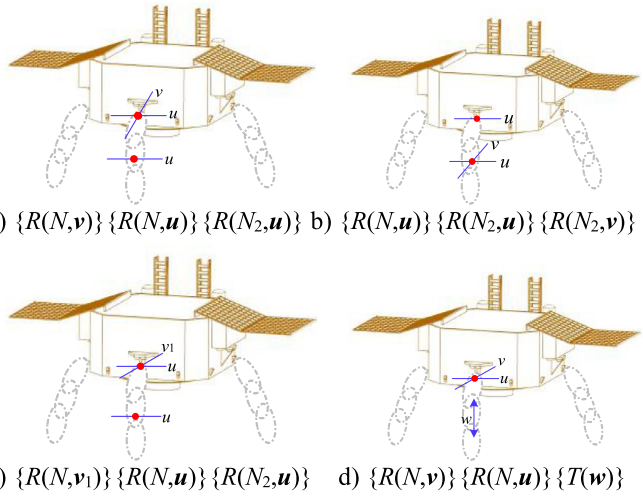


Fig. 3. Motion requirements of LMLs.

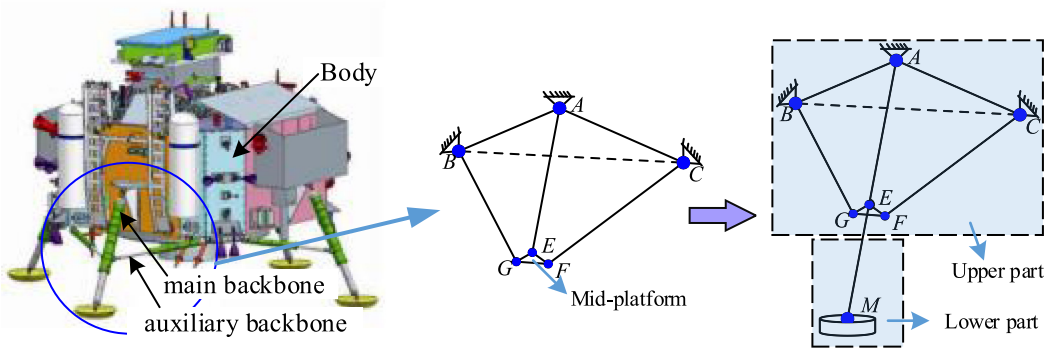


Fig. 4. Truss design of the leg for LMLs.

As for the upper part: The upper part is regarded as a parallel mechanism. Based on Eq. (4), the numbers could be achieved,

$$n_{\mathrm{AL}} = E_L = 3, n_{\mathrm{PL}}=0, n_{\mathrm{TL}}= 3 \tag{15}$$

Since the upper part of the leg is without redundant DoF, its DoF is equal to its dimension of motion,

$$M = M_{DoF} = 3 \tag{16}$$

For facilitating the designing of the mechanism, λ, v, and ξ is set to 0. There are two loops because n_{TL}=3. Thus, we can get the total number of DoF in jth limbs as follows:

$$\sum_{i=1}^{n_{TL}} C_i = M+dl= 3 + 6 \times 2=15 \tag{17}$$

According to the combination principle of parallel mechanism, the DoF of each limb is not less than that of the moving platform, detailed as following,

$$3 \le C_j \tag{18}$$

Since mechanisms without redundant DoF are yet to be designed, the DoF of each limb is not more than 6, which is

$$C_j \le 6 \tag{19}$$

Thus, it can be obtained as follows:

$$3 \le C_j \le 6 \tag{20}$$

There are three limbs; thus, a DoF can be allocated to each limb, which has three cases, listed in Table 2.

Table 2
Distribution of DoF.

Cases	C_i		
	C_1	C_2	C_3
Case 1	3	6	6
Case 2	4	5	6
Case 3	5	5	5

Table 3
Backbones with {*D*}.

Joints	Limbs		
P U S	UPS	PUS	PSU SPU
R U S	RUS	RSU	URS SRU
Pa U U* S	UPaS	PaUS	PU*S PaU*S

Table 4
Typical configurations of legs.

Main backbones	Auxiliary backbones			Transmission limbs
UR	2-RUS	2-PUS	2-PU*S	**RRR**
	2-URS	2-UPS	2-U*PS	PRR
UP	**2-RUS**	2-PUS	2-PU*S	RS**S**
	2-UPS	2-URS	2-U*PS	PSS
RU	**2-RUS**	2-PUS	2-PU*S	
	2-URS	2-UPS	2-U*PS	

According to Lie group theory, the distribution of DoFs (which is listed in Table 2), intersection rules, and corresponding procedure, all the displays of the LMCs can be easily obtained. Considering the scope of this study, only one display of LMCs without overconstraint (according to Eq. (11)) is presented, which belongs to Case 1 listed in Table 2. Thus, the layout of the submanifold of the limbs is chosen as $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}\&\{D\}\&\{D\}$, $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\&\{D\}\&\{D\}$, $\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\{R(N_2,\boldsymbol{v})\}\&\{D\}\&\{D\}$, $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\&\{D\}\&\{D\}$ and $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}\&\{D\}\&\{D\}$, as shown in Fig. 5.

For the main backbone: the motions of the main backbones are $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$, $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$, $\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\{R(N_2,\boldsymbol{v})\}$, $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$ and $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$. According to the Lie group theory, if the submanifold is $\{L\}=\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$, the corresponding structures with simple joint can be ${}^{v}_{N}\mathrm{R}^{u}_{N}\mathrm{R}^{w}\mathrm{P}$, ${}^{u}_{N}\mathrm{R}^{v}_{N}\mathrm{R}^{w}\mathrm{Pa}$. Furthermore, more structures with complex joints can be obtained by employing the complex joint to replace the simple joints, that are ${}^{vu}_{N}\mathrm{U}^{w}\mathrm{P}$ and ${}^{vu}_{N}\mathrm{U}^{w}\mathrm{Pa}$. Similarly the structures for $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$, $\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\{R(N_2,\boldsymbol{v})\}$, $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$ and $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$ are ${}^{vu}_{N}\mathrm{U}^{u}\mathrm{R}$, ${}^{u}\mathrm{R}^{uv}_{N}\mathrm{U}$, ${}^{w_1u}_{N}\mathrm{U}^{u}\mathrm{R}$, and ${}^{w_1u}_{N}\mathrm{U}^{w}\mathrm{P}$, respectively.

According to the methods to maintain TPM and the methods (explained in detail in Section 2) to arrange the joints to the truss, the joints are arranged into the main backbone of the truss to achieve the two types of arrangements. (1) As for the main backbones ${}^{vu}_{N}\mathrm{U}^{w}\mathrm{P}$, ${}^{vu}_{N}\mathrm{U}^{u}\mathrm{R}$, ${}^{w_1u}_{N}\mathrm{U}^{u}\mathrm{R}$ and ${}^{w_1u}_{N}\mathrm{U}^{w}\mathrm{P}$, P or R joint can be added between the node *A* and *E* and the U joint is attached in the node *A*. It can be noted that the directions of the two links attached to the R joint are coincident to maintain the singularity position of TPM. (2) For ${}^{u}\mathrm{R}^{uv}_{N}\mathrm{U}$, the R joint can be added before the link *AE*, and the U joint is attached in the node *A* to get TPM, as shown in Fig. 6(c). The five structures of the main backbones are shown in Fig. 6.

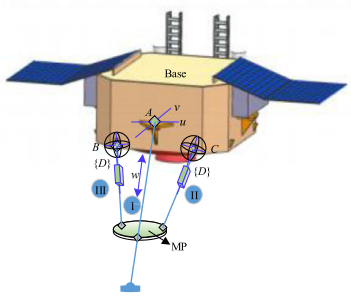
For the auxiliary backbone: the submanifold is

$$\{L_i\} = \{D\} = \{R(\boldsymbol{u})\}\{R(\boldsymbol{v})\}\{R(\boldsymbol{w})\}\{T(\boldsymbol{u})\}\{T(\boldsymbol{v})\}\{T(\boldsymbol{w})\} \tag{21}$$

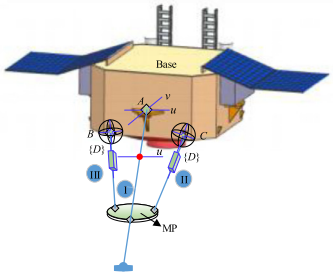
According to the product closure in subgroups, $R(N, x)$ and $T(\boldsymbol{x})$ ($\boldsymbol{x}=\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}$) are the displacement subgroups of {*D*}, and their product $R(N, \boldsymbol{x})$ and $T(\boldsymbol{x})$ can commute to each other. The mechanical generators of this manifold {*D*} are listed in Table 3.

For simplicity, we consider the RUS limb as an example to arrange the joints into the truss, and the joints of other limbs can be arranged in the same way. Similarly, the joints are arranged into the truss to achieve the three types of arrangements. (1) The R joint is attached in the node *B*, the U joint is added between the node *B* and *G*, and the S joint is added in the node *G*. It can be noted that the directions of the two links attached to the U joint are coincident to maintain the singularity position. (2) The R joint can be added before the link *BG*, the U joint is attached in the node *B* and fixed to get the TPM, and the S joint is added in the nod *G*. (3) The R joint is attached in the node *B*, the U joint is attached in the node *G* and fixed on the link *AE* to get the TPM, and the S joint is allocated on the extended link AE after RU link. Then, three types of auxiliary limbs are obtained, as shown in Fig. 7(a), (b) and (c), respectively.

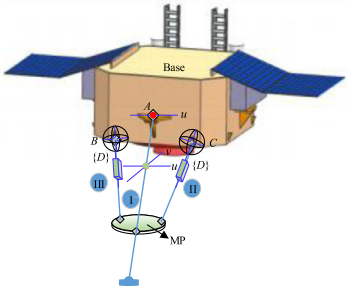
The structures of the legs are synthesized by assembling three corresponding backbones under the guidance of the topology of the truss. These typical configurations of the legs are listed in Table 4. Note that according to the structures



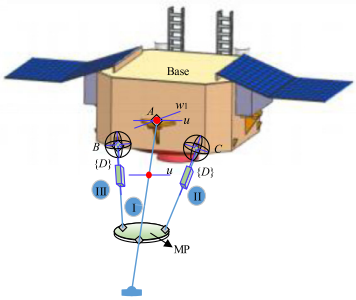
a) $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}\&\{D\}\&\{D\}$



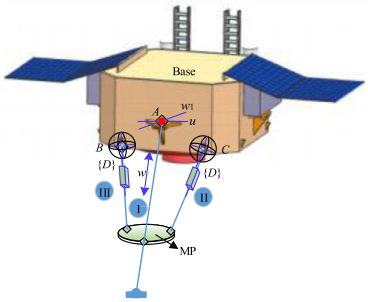
b) $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\&\{D\}\&\{D\}$



c) $\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\{R(N_2,\boldsymbol{v})\}\&\{D\}\&\{D\}$

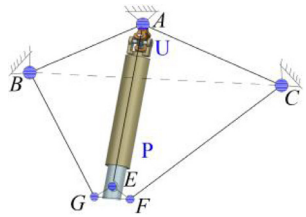


d) $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\&\{D\}\&\{D\}$

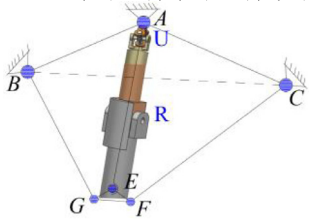


e) $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}\&\{D\}\&\{D\}$

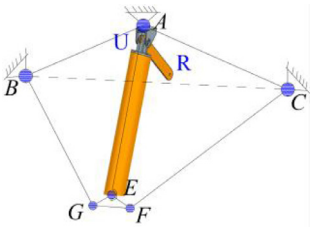
Fig. 5. Arrangements of the motions of the backbones.



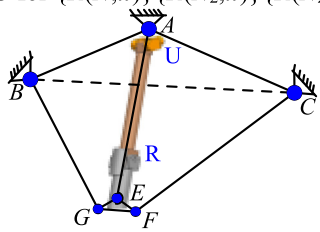
a) UP for $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$



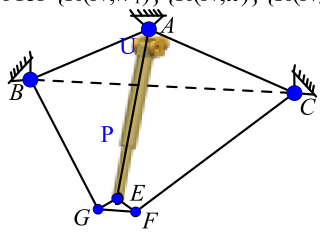
b) UR for $\{R(N,\boldsymbol{v})\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$



c) RU for $\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}\{R(N_2,\boldsymbol{v})\}$



d) UR for $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{R(N_2,\boldsymbol{u})\}$



e) UP for $\{R(N,\boldsymbol{w}_1)\}\{R(N,\boldsymbol{u})\}\{T(\boldsymbol{w})\}$

Fig. 6. Structures of the main backbones.

of auxiliary backbones, each structure has three types of configuration. Considering UP&2-RUS as an example, the three typical configurations during the landing phase are shown in Fig. 8(a), (b) and (d), which are called as case I, case II and case III mechanisms, respectively. During the landing phase, for case I, the directions of the two links of the same auxiliary backbone are collinear to reach the TPM-DPB. However, for case II, during the landing phase, the node *B*/*C* (or the U joint) of the auxiliary backbones are fixed by electromagnetic bolts to reach the TPM-FJB. For case III, the node *F*/*G* (or the U joint) of the auxiliary backbones are fixed by electromagnetic bolts to reach the TPM-FJB.

Note that the main and auxiliary backbones can be replaced by other corresponding backbones at the same TPM, listed in Table 4. For example, consider RU&2-RUS as an example, which is case III, as shown in Fig. 8(c). For case III, during the landing phase, the nodes A, B, and C (or the U joint) of the main and auxiliary backbones are fixed on the body of LMLs by electromagnetic bolts to reach the TPM to bearing the large impact. Fig. 8(d) illustrates another type of RU&2-RUS mechanism, which is case IV. The U joints of the auxiliary backbones are fixed on the main backbone by the electromagnetic

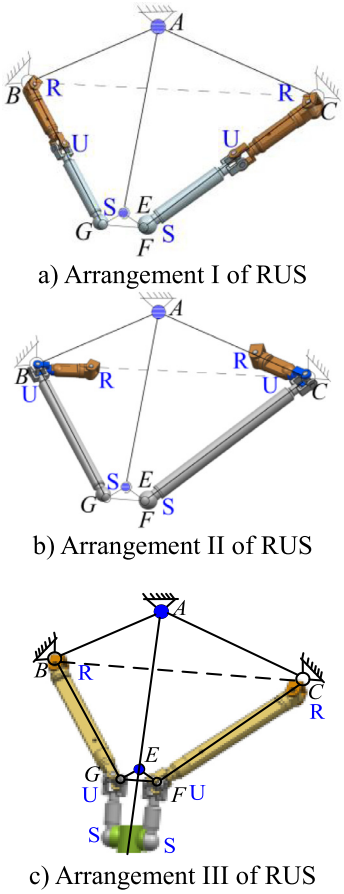


Fig. 7. Structures of the auxiliary backbones.

bolts, while the U joint of the main backbone is fixed on the body. Furthermore, a transmission limb is added in each auxiliary backbone.

As for selection of the actuated joints and the transmission limb: according to the principle stated in Section 3.4, the rules to select the actuated joints for these mobile landers are as follows:

Rule 1: Only one joint is actuated in each limb.

Rule 2: The joint connected between the link and the base is enacted as an actuated joint, preferably to reduce the inertia.

Rule 3 (for the main backbone): If the main backbone is UP limb, the P joint is selected as the actuated joint; if the main backbone is RU limb, the R joint is selected as the actuated joint.

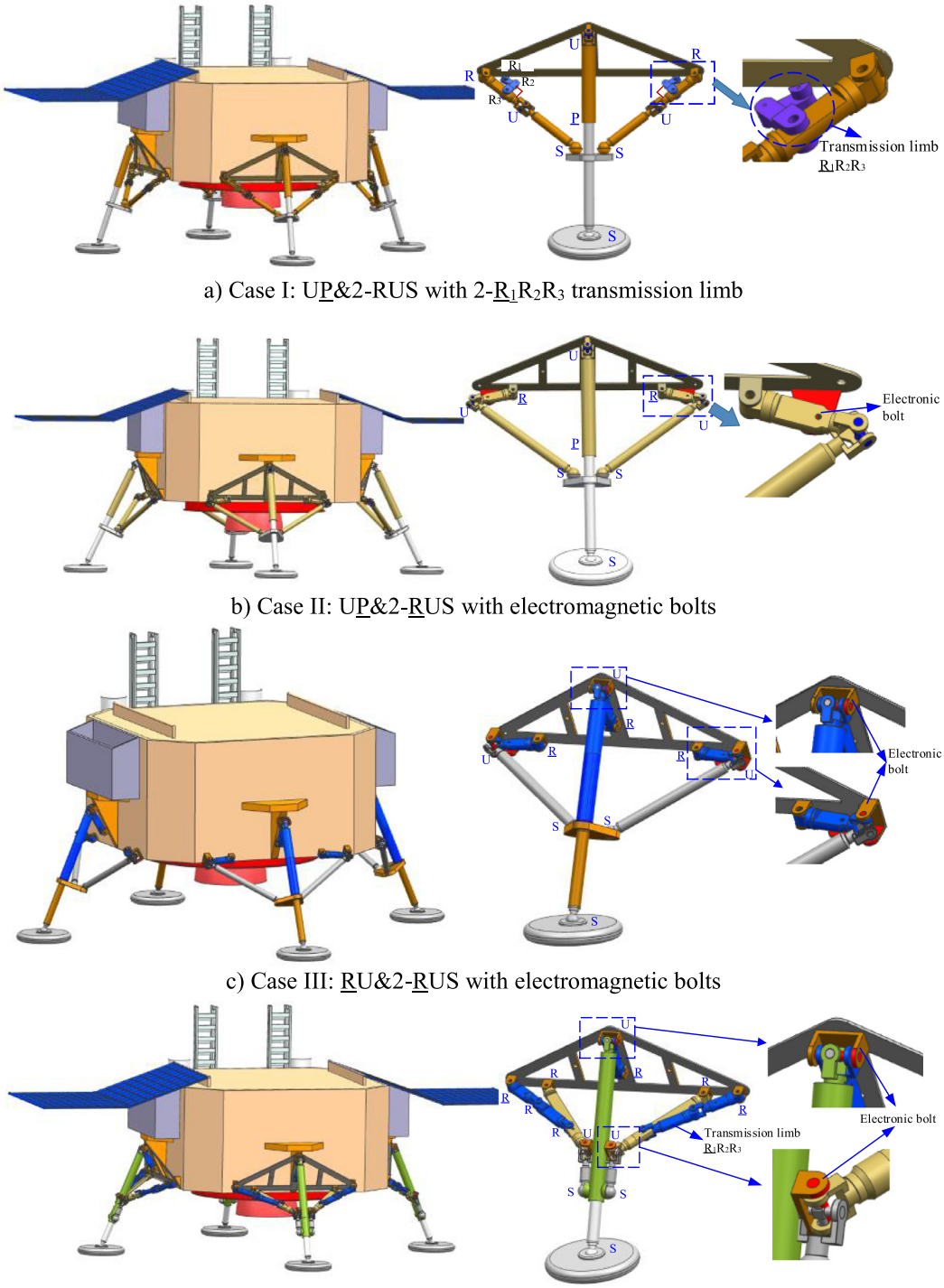
Rule 4 (for the auxiliary backbones): As for the TPM 1 (such as the auxiliary limbs of Case I mechanism), to improve the performance of the robot, a transmission limb is added to actuate one joint in each auxiliary backbone. Accordingly, the 1D joint needs to be actuated. The transmission limb could be RRR, PRR, RSS, or PSS. According to rule 1, the first joint attached to the base of the transmission is the actuated joint, as shown in Fig. 8(a). For TPM 2 (such as the auxiliary limbs of Case II mechanism), the first joint attached to the base is selected as the actuated joint, as shown in Fig. 8(b).

Rule 5, for the lower part, the S joint is designed as a passive joint by adding some particular springs in order to improve the terrain adaptability. The symbols of the actuated joints are represented with an underscore, such as $\underline{P}$ or $\underline{R}$, as shown in Fig. 8.

4.4. Type evaluation of legs for LMLs

Based on the above-mentioned two criterions, the mechanisms with better performance are selected.

Consequently, the number of overconstraints of the four cases were obtained according to Eqs. (12) and (13), i.e., $\Delta 1=\Delta 2=\Delta 3=\Delta 4=3-3=0$. The number of overconstraints equals zero. Furthermore, for these four cases of mechanisms, all the actuated joints connected to the base are selected. Thus, the three cases are the proper candidates for LML legs.



a) Case I: UP&2-RUS with 2-$\underline{R}_1R_2R_3$ transmission limb

b) Case II: UP&2-RUS with electromagnetic bolts

c) Case III: RU&2-RUS with electromagnetic bolts

d) Case IV: RU&2-RUS with electromagnetic bolts and 2-$\underline{R}_1R_2R_3$ transmission limb

Fig. 8. Three typical structures of LMLs.

5. Two typical LMLs analyses

Numerous LMLs with structurally symmetric legs are obtained by assembling a body with the same structures of legs listed in Table 4. Two typical cases of LMLs (whose mechanisms of the legs are presented in bold and listed in Table 4) are taken as examples (shown in Fig. 8) to verify the multiple functions of LMLs including being deployable, landing buffer,

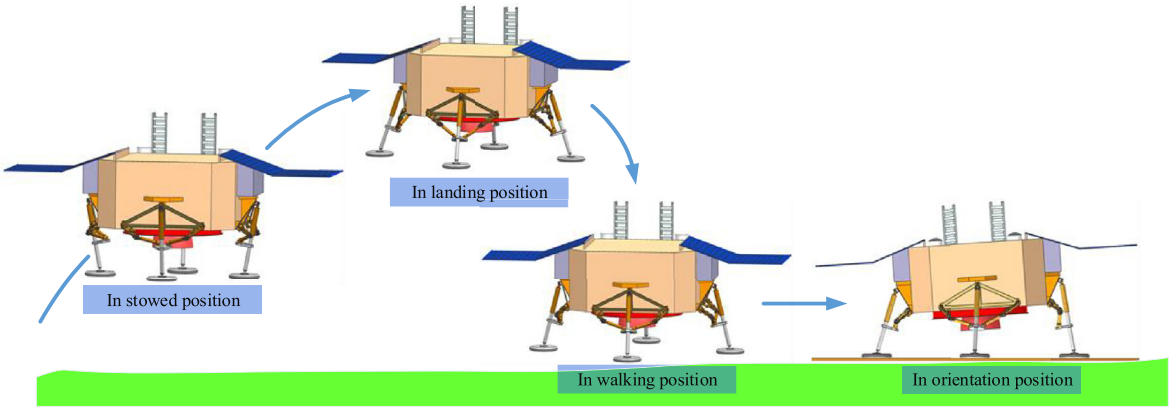


Fig. 9. Lander's positions in different phases.

walking, orientation adjustment, and terrain adaptability. Furthermore, other configurations of LMLs can be verified by the same method.

Fig. 8(a) shows one typical mechanism (Type I) of LMLs' legs, which is presented as (UP&2-RUS)-S. The three configurations including the stowed position, landing position and walking and orientation positions are explained in detail as follows. (1) In the stowed position, each leg rotates around the axis of the first joints (U joints) of the main backbone to reach the stowed status by actuating its two actuated joints in auxiliary backbones. One locking and clutch device is designed in each main backbone to lock the P joint. The structure of the upper part legs can be presented as U&2-RUS with RRR transmission limbs, and the motion of its mid-platform is 2-D rotations. Further, the LML changes its configuration from the stowed position to the deployed position by actuating the three actuated joints and unlocking the P joint of the main backbone. (2) In its landing position (the deployed position), the directions of the two links of the same auxiliary backbone are collinear to reach the TPM-DPB and minimize shock damage during landing. The structure of this LML can be represented as 4-{(U&2-RS)-S} with RRR transmission limbs, and the motion of its body is {*E*} which means it has no motion. Furthermore, the axes of the R joints of the auxiliary backbone are perpendicular to the directions of the adjacent links in the same limbs. The R_2R_3 link of the transmission limb are perpendicular to the axis of the auxiliary limb. Thus, the actuators in the transmission limb will not bear any impact force, i.e., the actuators are protected from the impact force. (3) In its walking position, after landing, the configuration of the mobile lander is changed from the landing status to the walking status by actuating the three actuated joints and unlocking the P joint of the main backbone. The directions of the two links of the auxiliary backbones are not collinear. The structure of this LML can be presented as 4-{(UP&2-RUS)-S} with RRR transmission limbs, and the motions of its body and the foot pad are {*D*}, which means the LML has 6-D motions with flexible ability to walk. (4) In the orientation position, the lander can be regarded as a parallel mechanism and the body is the moving platform, whose limbs are its legs that are in contact with the ground, which is also presented as 4-{(UP&2-RUS)-S} with RRR transmission limbs. The body's motion is {*D*} because the motion of the legs is {*D*}, which indicates that the body can rotate around any axis and translate in any direction, which means it exhibits orientation adjustment. The positions of the LML with different status are shown in Fig. 9.

Fig. 8(c) shows another typical mechanism (Type III) of LMLs' legs, which is presented as (RU&2-RUS)-S. For this type, the four positions are explained in detail as follows: (1) in the stowed position, the RU link of the main backbone is fixed by the electronic bolts, and the leg rotates around the axis of the U joints of the main backbone to reach the stowed status by actuating the two actuated joints in auxiliary backbones. The structure of the upper part legs can be presented as U&2-RUS, and the motion of its mid-platform has 2-D rotations. Further, the LML changes its configuration from the stowed position to the deployed position by actuating the two actuated joints in the auxiliary backbones. (2) In its landing position (the deployed position), one side of the US links of the auxiliary backbones is fixed by the electronic bolts that are installed on the body of LMLs. The structure of this LML can be presented as 4-{(U&2-US)-S}, and the motion of its body is {*E*} which means it has no motion. Furthermore, the three actuators are protected from the impact force by fixing the RU links in order that the impact force does not pass through the RU links. (3) In its walking position, after landing, the configuration of the LML is transformed from the landing position to the walking position by actuating the three actuated joints and unlocking the electronic bolts in each backbone. The structure of this LML can be presented as 4-{(RU&2-RUS)-S}, and the motions of its body and the foot pad are {*D*}, which means the LML has 6-D motions with flexible ability to walk. (4) In the orientation position, its configuration is the same as that in the walking position with four legs standing on the ground. In this position, the lander can be regarded as a parallel mechanism and the body is the moving platform, whose limbs are its legs that are in contact with the ground, which is also presented as 4-{(RU&2-RUS)-S}. The body's motion is {*D*} because the motion of the legs is {*D*}, which means the body can rotate around any axis and translate in any direction, which means it exhibits orientation adjustment.

6. Conclusion

LSLs have made a significant contribution to soft landing and exploration on the extraterrestrial body's surface. To expand their detection range further, and meet the requirements of base construction in the future, we propose a novel LML concept. To realize the transformation of its landing and walking configurations, the TMT method is proposed in this study for designing LMLs.

In terms of design approach, we investigated an approach of type synthesis of mechanisms (TMT method). This approach has two advantages. First, it is appropriate to design robots with multiple functions, such as landing and walking. Second, the obtained mechanisms can bear a larger impact or load when the mechanism located TPM because it can be regarded as a structure or truss. The TMT method includes the following four stages: (1) obtain the motion requirements, (2) design trusses for the task, (3) motion design of a mechanism based on the truss and the motion requirements, and (4) perform type evaluation of the corresponding mechanisms. The TMT method provides a new method of designing robots with the capabilities to transform its kinematics from a truss to a mechanism and vice-versa.

In terms of application, numerous novel LMLs have been proposed that have the inbuilt features of multiple functions including landing and walking capabilities, as well as being deployable, orientation adjustable, and terrain adaptable. During the landing phase, for the Type I lander, all the limbs of the LMLs' legs are in their TPM-DPB to protect their joints and actuators; for the Type II, Type III and Type IV LMLs, the actuators of the auxiliary backbone are protected from the compact force in the TPM-FJB, which is realized by using electronic bolts. Based on the TMT method, four typical novel LMLs are obtained, which can extend the range of exploration. Furthermore, the LML structures provide some technical support and mechanisms for deep space exploration.

Declaration of Competing Interest

None.

Acknowledgments

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